

Mitigating Task Interference in Multimodal Glioma Segmentation via Region-Specific Modality Gating

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Abstract—Multimodal MRI glioma segmentation requires different modality cues for enhancing tumor (ET), tumor core (TC), and whole tumor (WT). Existing shared-fusion and global-gating methods often apply one modality policy to all sub-regions, which can introduce task interference. We propose a region-specific modality gating framework that learns separate gates for ET, TC, and WT within a hybrid CNN-Transformer architecture. The model dynamically weights T1, T1Gd, T2, and FLAIR for each sub-region, while Grad-CAM-guided distillation and sparsity regularization encourage stable and interpretable modality selection. On BraTS 2019, the proposed method achieves Dice scores of 0.914/0.928/0.921 and HD95 values of 4.98/6.72/5.43 mm for ET/TC/WT, outperforming shared-fusion and global-gating baselines. Consistent gains on BraTS 2021 suggest improved generalization. These results indicate that region-specific gating improves accuracy, robustness, and interpretability in multimodal glioma segmentation.

Index Terms—Multimodal glioma segmentation, magnetic resonance imaging (MRI), region-specific gating, task-aware fusion, multimodal learning, CNN-Transformer, medical image segmentation.

I. INTRODUCTION

Adult gliomas are primary brain tumors that exhibit substantial variation in morphology, location, growth pattern, and malignancy, making accurate delineation of tumor sub-regions essential for treatment planning and radiation targeting [1], [2]. In clinical practice, three sub-regions of particular importance are enhancing tumor (ET), tumor core (TC), and whole tumor (WT). Magnetic resonance imaging (MRI) is the clinical standard for glioma assessment as it provides complementary information through T1, T1 with gadolinium contrast (T1Gd), T2, and fluid-attenuated inversion recovery (FLAIR). These sequences capture different biological and structural properties, where T1 provides anatomical detail, T1Gd highlights enhancing tumor tissue, T2 emphasizes fluid-sensitive abnormalities, and FLAIR reveals edema and whole-tumor extent [2], [3]. Since no single sequence fully characterizes glioma appearance, accurate segmentation depends on effective multimodal fusion.

In practice, the three glioma sub-regions exhibit distinct radiological characteristics and depend on different modality cues. When a model uses one shared multimodal representation for all three sub-regions, modality preferences that benefit one target can suppress or degrade information needed by

another. This negative transfer, or task interference, can reduce region-wise segmentation accuracy and limit the interpretability of multimodal fusion.

Although attention-based fusion methods have improved patient-specific modality weighting, there is still room for improvement in task-aware multimodal glioma segmentation. Existing global fusion or global gating strategies do not explicitly account for the distinct imaging requirements of ET, TC, and WT. As a result, a single shared gate may converge to an average modality configuration that is suboptimal for individual sub-regions. To address this limitation, we present a region-specific modality gating framework for mitigating task interference in multimodal glioma segmentation. Our method learns separate modality gate vectors, $\mathbf{z}^{\text{ET}}$, $\mathbf{z}^{\text{TC}}$, and $\mathbf{z}^{\text{WT}}$, allowing each sub-region to dynamically emphasize informative MRI modalities while suppressing irrelevant or noisy inputs.

The proposed approach is built on a hybrid CNN-Transformer architecture with modality-specific feature streams and region-aware gating. The gating module is conditioned on modality-specific features and fused contextual representations to produce sub-region-specific modality weights. To encourage stable and interpretable gating behavior, we incorporate attention-guided and Grad-CAM-guided distillation together with sparsity regularization. This promotes efficient, clinically meaningful modality selection [17].

The primary objective of this study is to improve the accuracy, interpretability, and robustness of joint ET/TC/WT segmentation by reducing task interference through region-specific modality gating. To evaluate the proposed framework, we utilize the BraTS 2019 and BraTS 2021 adult glioma datasets and analyze region-wise performance, modality selection behavior, and robustness under varying modality conditions [2], [18]. By decoupling modality usage across ET, TC, and WT, the proposed approach is designed to provide more reliable multimodal segmentation for clinically relevant glioma sub-region analysis. RSMG achieves the highest Dice scores among the compared BraTS 2019 methods and consistently outperforms shared-fusion and global-gating baselines on BraTS 2019 and BraTS 2021, supporting region-specific modality gating as an effective strategy for accurate, inter-

pretable, and robust glioma segmentation.

Our main contributions can be summarized as follows:

- 1) We introduce a novel region-specific modality gating framework that learns distinct modality combinations for ET, TC, and WT in multimodal glioma segmentation.
- 2) We formulate task-aware multimodal fusion within a hybrid CNN–Transformer pipeline to mitigate task interference in joint sub-region prediction.
- 3) We incorporate attention-guided and Grad-CAM-guided distillation together with sparsity regularization to promote stable, interpretable, and clinically meaningful modality selection.
- 4) We design a comprehensive evaluation on BraTS 2019 and BraTS 2021 to assess region-wise segmentation performance, gating behavior, and robustness under varying modality conditions.

II. RELATED WORK

CNN and hybrid segmentation architectures have become widely used for glioma segmentation in multimodal MRI, supported by benchmark datasets such as BraTS [2], [18]. CNN encoder–decoder models such as U-Net and 3D U-Net established strong baselines for voxel-wise medical image segmentation [4], [5]. More recently, hybrid CNN–Transformer architectures have incorporated self-attention to model long-range context while retaining CNN-based locality [6], [7]. Even with strong segmentation accuracy, many multimodal approaches still concatenate modalities at the input stage or fuse them using a shared strategy, which limits patient-specific and region-specific modeling of modality relevance.

Multimodal fusion and interpretability are increasingly important because different MRI sequences provide different information for glioma sub-regions. T1 provides anatomical detail, T1Gd highlights enhancing tumor tissue, T2 emphasizes fluid-sensitive abnormalities, and FLAIR reveals edema and whole-tumor extent [2], [3]. Explainable multimodal architectures and saliency-based methods such as Grad-CAM can help visualize the image regions and modality cues that influence model predictions [8], [17]. However, many existing fusion methods estimate modality importance globally or after feature extraction, so the learned weighting does not explicitly separate the needs of ET, TC, and WT.

Segmentation with missing modalities has also been extensively studied in multimodal brain tumor MRI. Recent methods have explored transformer-based modeling, modality compensation, masked pretraining, state-space fusion, masked mutual learning, and knowledge-distillation-based adaptation to preserve accuracy when one or more MRI sequences are unavailable [9]–[16]. While these methods improve robustness to incomplete inputs, they primarily focus on missing-modality adaptation rather than the distinct modality requirements of individual tumor sub-regions.

Region-specific modality modeling remains less explored in multimodal glioma segmentation. In practice, ET is typically most visible in T1Gd, WT and edema are better characterized by T2 and FLAIR, and TC benefits from both structural and

contrast information [8]. A single shared fusion or global gating strategy can therefore produce an average modality configuration that is suboptimal for individual regions. RSMG addresses this limitation by learning separate modality gates for ET, TC, and WT, enabling task-aware fusion that reduces interference while improving interpretability and robustness.

III. METHODOLOGY

A. Dataset and Pre-processing

We evaluate the proposed framework on the BraTS 2019 and BraTS 2021 adult glioma datasets [2], [18]. Each subject includes four co-registered MRI modalities (T1, T1Gd, T2, and FLAIR) along with expert annotations for enhancing tumor (ET), tumor core (TC), and whole tumor (WT).

All MRI volumes are pre-processed using the same pipeline as the backbone segmentation model. Specifically, non-brain tissues are removed using skull stripping, the four modalities are spatially aligned, volumes are resampled to $1 \times 1 \times 1$ mm³, and each modality is normalized independently using Z-score normalization. Non-brain voxels are set to zero, and each volume is cropped to a fixed in-plane size of 192×192 to reduce unnecessary computation. The patients were partitioned into five non-overlapping folds. For each cross-validation run, one fold was held out as the test set, and the remaining four folds were used for model development. Performance was computed on the held-out test fold for each run, and we report the mean performance across the five test folds.

B. Backbone Architecture

Our method is built on a hybrid CNN–Transformer backbone with modality-specific feature extraction and adaptive multimodal fusion [19], [20]. Fig. 1 provides an overview of the proposed framework, including modality-specific CNN encoders, cross-modal interaction, region-specific gating, a shared CNN decoder, and separate ET, TC, and WT prediction heads.

Let $M = 4$ denote the number of MRI modalities. For each modality $m \in \{1, \dots, M\}$, a modality-specific 3D CNN encoder produces a feature map F_m . Each encoder consists of 3D convolutional blocks with feature depths of 16, 32, 64, and 128, followed by batch normalization and ReLU activations. The encoder outputs are partitioned into non-overlapping $16 \times 16 \times 16$ patches, linearly projected into embeddings, and processed by a modality-specific Transformer encoder with four layers, eight attention heads, and a 512-unit MLP.

After modality-specific encoding, adaptive self-attention produces modality-aware features $\tilde{F}_m$. The backbone then computes global modality-importance weights

$$\beta = \text{Softmax}\left([\text{GAP}(\tilde{F}_1), \dots, \text{GAP}(\tilde{F}_M)] W_\beta \right), \quad (1)$$

where $\text{GAP}(\cdot)$ denotes global average pooling and W_β is a learned projection. The fused multimodal representation is

$$F_{\text{fused}} = \sum_{m=1}^M \beta_m \tilde{F}_m. \quad (2)$$

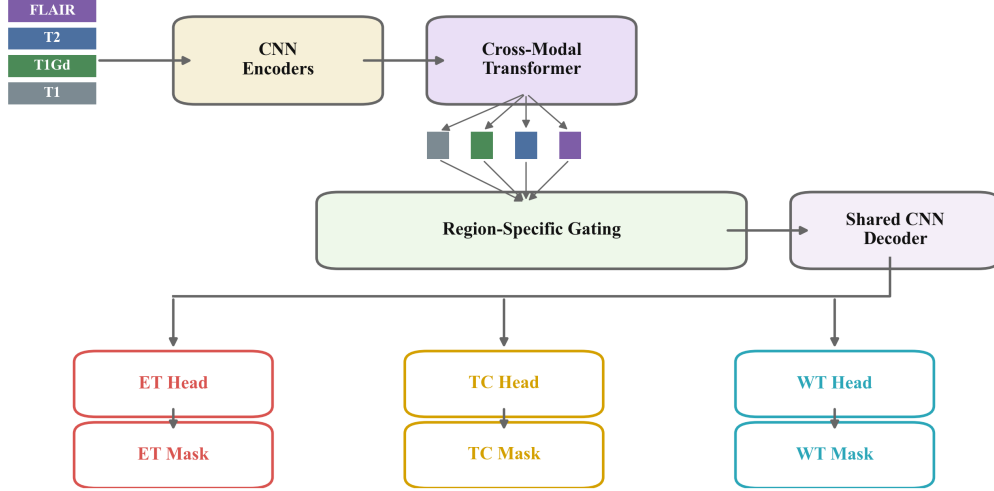


Fig. 1. Overview of the proposed region-specific gating framework. Four MRI modalities (T1, T1Gd, T2, and FLAIR) are encoded by modality-specific CNN encoders and processed by a cross-modal transformer that preserves modality-aware feature streams. Region-specific gates z^{ET} , z^{TC} , and z^{WT} construct ET, TC, and WT specific fused features, which are refined by a shared CNN decoder and mapped to binary masks.

Unlike the backbone, which uses a single shared multimodal representation for all outputs, the proposed method introduces task-aware region-specific gating before ET, TC, and WT prediction.

C. Region-Specific Modality Gating

The key idea of the proposed framework is to replace a single global modality strategy with separate modality gates for each tumor sub-region. Let

$$\mathbf{z}^r = [z_1^r, z_2^r, \dots, z_M^r] \in [0, 1]^M, \quad r \in \{ET, TC, WT\}, \quad (3)$$

where z_m^r controls the contribution of modality m to region r . In this way, ET, TC, and WT are allowed to rely on different modality combinations [19], [20].

To predict the gates, we first summarize each modality-specific feature stream and the fused representation using global average pooling. Here, $\phi^r = \text{GAP}(H^r) \in \mathbb{R}^{C_h}$ is obtained by global average pooling the feature map H^r of the corresponding region head h_r , where C_h is the number of head channels and gradients are allowed to propagate through this descriptor during training. The gating logits for region r are then obtained by a lightweight MLP:

$$\mathbf{a}^r = g_r \left([\text{GAP}(\tilde{F}_1), \dots, \text{GAP}(\tilde{F}_M), \text{GAP}(F_{\text{fused}}), \phi^r] \right), \quad (4)$$

where $g_r(\cdot)$ is a region-specific gating network.

During training, we use a differentiable relaxation so that the gates remain trainable end-to-end [21], [22]:

$$\tilde{z}_m^r = \sigma \left(\frac{a_m^r + \epsilon_m^r}{\tau} \right), \quad (5)$$

where $\sigma(\cdot)$ is the sigmoid function, τ is a temperature parameter, and ϵ_m^r is injected noise used to encourage near-binary decisions. At inference time, the gates can be used

directly as soft weights or hardened by thresholding for more interpretable modality selection.

The region-specific feature representation is then computed as

$$F^r = \sum_{m=1}^M \tilde{z}_m^r \tilde{F}_m, \quad (6)$$

and each region head predicts a binary mask

$$\hat{Y}^r = h_r(F^r), \quad (7)$$

where $h_r(\cdot)$ denotes the prediction head for region r . Because the three heads no longer share a single modality policy, the proposed formulation reduces negative transfer caused by forcing ET, TC, and WT to use the same multimodal configuration.

D. Gate Supervision and Optimization

Each region head is supervised using its own segmentation target. Let Y^r and $\hat{Y}^r$ denote the ground-truth and predicted masks for region r , respectively. The region-wise segmentation loss is defined as

$$\mathcal{L}_{\text{seg}} = \sum_{r \in \{ET, TC, WT\}} \left[1 - \text{DSC}(\hat{Y}^r, Y^r) + \lambda_{\text{ce}} \text{CE}(\hat{Y}^r, Y^r) \right], \quad (8)$$

where $\text{DSC}(\cdot, \cdot)$ is the Dice similarity coefficient and $\text{CE}(\cdot, \cdot)$ is the cross-entropy loss.

To encourage clinically meaningful modality usage, we derive a target modality-importance distribution for each region from Grad-CAM activations computed within the corresponding tumor mask [17]. Specifically, for each region r and modality m , Grad-CAM activations are computed from the region-specific logit of head h_r with respect to the modality feature map $\tilde{F}_m$, averaged within the ground-truth mask Y^r ,

and normalized across the four modalities to obtain $q^r \in \mathbb{R}^M$. Let $\mathbf{q}^r$ denote the normalized target distribution for region r , and let $\mathbf{p}^r$ denote the normalized predicted gate distribution derived from $\tilde{\mathbf{z}}^r$. The distillation objective is

$$\mathcal{L}_{\text{distill}} = \sum_{r \in \{\text{ET}, \text{TC}, \text{WT}\}} \text{KL}(\mathbf{q}^r \parallel \mathbf{p}^r), \quad (9)$$

which guides the gates toward region-consistent modality patterns while still allowing unhelpful modalities to be down-weighted [23].

To keep the selected modality subsets compact, we add a sparsity penalty

$$\mathcal{L}_{\text{sparse}} = \sum_{r \in \{\text{ET}, \text{TC}, \text{WT}\}} \alpha_r \|\tilde{\mathbf{z}}^r\|_1, \quad (10)$$

where α_r controls the sparsity level for each region. The final objective is

$$\mathcal{L} = \mathcal{L}_{\text{seg}} + \lambda_d \mathcal{L}_{\text{distill}} + \lambda_s \mathcal{L}_{\text{sparse}}. \quad (11)$$

E. Training Setup and Evaluation Protocol

We employ a two-stage training strategy. In the first stage, the multimodal backbone is trained using all four MRI modalities to learn stable shared and modality-specific features. In the second stage, the region-specific gating module is initialized on top of the pretrained backbone and trained jointly with the segmentation heads, while the backbone is fine-tuned with a smaller learning rate for stability.

The backbone is optimized using stochastic gradient descent with momentum 0.9, an initial learning rate of 8×10^{-3} , batch size 16, and 250 epochs per fold. Gate-related hyperparameters, including λ_d , λ_s , λ_{ce} , and the temperature schedule τ , are selected on the validation split. We compare the proposed method against two baselines: (1) the shared-fusion backbone without region-specific gating, (2) a global-gating baseline that uses one modality gate for all sub-regions.

Performance is measured using Dice and the 95th-percentile Hausdorff distance (HD95) for ET, TC, and WT. In addition to segmentation performance, we analyze the learned modality subsets for each region and evaluate robustness under missing-modality conditions by removing individual MRI sequences at test time [19], [24].

IV. RESULTS

We conducted a comprehensive evaluation of the proposed region-specific modality gating (RSMG) framework on the BraTS 2019 and BraTS 2021 datasets. In addition to within-study comparisons against two baselines, we also compared RSMG against reported multimodal glioma segmentation methods on BraTS 2019. The within-study baselines were: (1) a shared-fusion backbone that uses the same multimodal representation for ET, TC, and WT, and (2) a global-gating variant that learns a single modality gate shared across all three sub-regions. Performance was measured using Dice and the 95th-percentile Hausdorff distance (HD95) for ET, TC, and WT. In addition, we analyzed learned gate patterns,

TABLE I
COMPARISON WITH REPORTED MULTIMODAL GLIOMA SEGMENTATION METHODS ON BRA TS 2019 [2]. VALUES REPRESENT MEAN DICE SCORES.

Method	WT Dice $\uparrow$	TC Dice $\uparrow$	ET Dice $\uparrow$
DA-TransUNet [26]	0.830	0.810	0.750
MM-UNet [27]	0.870	0.790	0.740
MS U-Net [28]	0.880	0.800	0.760
DCMHA [29]	0.900	0.830	0.780
TransUNet 2D [6]	0.910	0.900	0.890
Proposed RSMG	0.921	0.928	0.914

qualitative segmentation behavior, robustness under missing-modality conditions, and the contribution of key components in the proposed model.

A. Comparison with Reported Methods

Table I provides a contextual comparison with reported multimodal glioma segmentation methods on BraTS 2019. The proposed RSMG achieves the highest Dice scores among the methods listed, with 0.914, 0.928, and 0.921 for ET, TC, and WT. These results are consistent with the direct within-study comparison in Table II, where RSMG also improves upon both the shared-fusion and global-gating baselines on BraTS 2019 and BraTS 2021. Together, these results support the claim of this work that learning separate modality policies for ET, TC, and WT reduces task interference more effectively than a single shared multimodal strategy.

B. Region-Specific Modality Patterns

To examine whether the learned gates capture meaningful modality preferences, we analyzed the learned region-specific modality weights on the BraTS 2019 test set. The ET head assigns its highest weight to T1Gd, consistent with the role of contrast enhancement in identifying active tumor. The WT head relies primarily on T2 and FLAIR, which better delineate edema and tumor extent. The TC head uses a more balanced combination of T1 and T1Gd, reflecting the need for both structural and contrast information. These findings indicate that the proposed gating module learns clinically meaningful and task-aware modality usage rather than consolidating to a single shared modality configuration [20].

C. Model Performance Evaluation

We compared the proposed model with the shared-fusion and global-gating baselines on BraTS 2019 and BraTS 2021. Table II summarizes Dice and HD95 results. The proposed RSMG model consistently outperforms both baselines across ET, TC, and WT on both datasets. On BraTS 2019, RSMG improves Dice from 0.901/0.916/0.909 to 0.914/0.928/0.921 and reduces HD95 from 5.41/7.26/5.88 mm to 4.98/6.72/5.43 mm relative to the shared-fusion baseline. This corresponds to an average Dice gain of approximately 1.2 percentage points and an average HD95 reduction of 0.47 mm. Similar gains are observed on BraTS 2021, where the proposed method improves Dice from 0.886/0.904/0.899 to 0.898/0.915/0.910 and reduces HD95 from 5.87/7.71/6.24 mm to 5.42/7.18/5.86 mm, suggesting improved generalization under cohort variability.

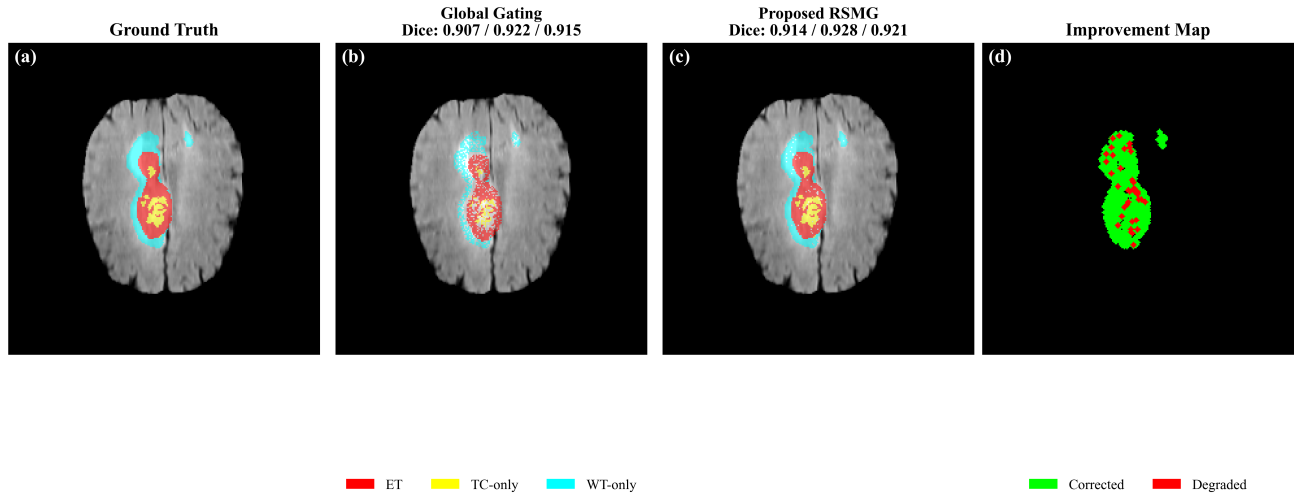


Fig. 2. Qualitative comparison of segmentation results. (a) Ground truth overlaid on FLAIR. (b) Global-gating baseline. (c) Proposed RSMG, which better preserves ET structure and reduces WT false positives. (d) Improvement map, where green indicates corrected regions and red indicates degraded regions.

TABLE II
SEGMENTATION RESULTS ON BRATS 2019 AND BRATS 2021.

Dataset	Method	Dice $\uparrow$			HD95 (mm) $\downarrow$		
		ET	TC	WT	ET	TC	WT
BraTS 2019	Shared fusion	0.901	0.916	0.909	5.41	7.26	5.88
	Global gating	0.907	0.922	0.915	5.20	6.98	5.61
	Proposed RSMG	0.914	0.928	0.921	4.98	6.72	5.43
BraTS 2021	Shared fusion	0.886	0.904	0.899	5.87	7.71	6.24
	Global gating	0.892	0.909	0.904	5.60	7.39	6.02
	Proposed RSMG	0.898	0.915	0.910	5.42	7.18	5.86

TABLE III
ABLATION RESULTS ON BRATS 2019.

Setting	Dice (ET/TC/WT) $\uparrow$	Avg. Active Modalities
Global gating baseline	0.907/0.922/0.915	2.9
w/o distillation	0.910/0.924/0.918	2.6
w/o sparsity	0.911/0.925/0.919	3.8
Proposed RSMG	0.914/0.928/0.921	2.3

D. Qualitative and Robustness Analysis

As shown in Fig. 2, qualitative analysis indicates that the proposed method provides cleaner ET boundaries, fewer false positives in edema-dominant WT regions, and improved preservation of TC structure in heterogeneous tumors. Grad-CAM analysis further indicated that the ET head concentrated more strongly on contrast-enhancing regions in T1Gd, while the WT head emphasized edema patterns in FLAIR and T2. This behavior is consistent with the learned region-specific modality weights and supports the interpretability of the proposed framework.

We also evaluated robustness under missing-modality conditions. When T1Gd was removed at test time, the proposed RSMG model achieved Dice scores of 0.782/0.857/0.878 for ET/TC/WT, compared with 0.761/0.846/0.871 for the global-gating baseline. When FLAIR was removed, RSMG achieved Dice scores of 0.915/0.888/0.845 versus 0.909/0.876/0.829 for global gating. The slight ET increase after removing FLAIR is marginal and should be interpreted as test-set variability rather than evidence that FLAIR improves ET segmentation, while the larger WT degradation is consistent with the stronger role of FLAIR in capturing edema and whole-tumor extent. These results suggest that region-specific gating can reallocate modality emphasis when an important sequence is unavailable, improving robustness to incomplete clinical inputs.

E. Ablation Study

To assess the contribution of each component, we performed an ablation study on BraTS 2019. Table III summarizes the ablation results. Removing region-specific specialization and reverting to a global gate reduces Dice across all three sub-regions. Excluding the distillation loss leads to weaker performance and less stable modality usage across folds, while removing sparsity increases the average number of active modalities and weakens the efficiency of the learned gates. The full model achieves the best balance between segmentation accuracy and efficient, interpretable modality selection.

These ablations indicate that region-specific specialization is the primary driver of the proposed gains. Compared with the full RSMG model, reverting to a single global gate reduces Dice from 0.914/0.928/0.921 to 0.907/0.922/0.915, suggesting that a shared modality policy cannot adequately capture the distinct imaging requirements of ET, TC, and WT. In contrast, removing sparsity increases the average number of active modalities from 2.3 to 3.8 while slightly reducing Dice, which shows that sparsity regularization encourages efficient modality selection without sacrificing segmentation accuracy. Together, these results show that the three components provide complementary benefits: region-specific gating primarily drives segmentation accuracy, distillation stabilizes learning, and sparsity promotes efficient modality selection.

Overall, the proposed region-specific modality gating framework improves region-wise segmentation accuracy, reduces

task interference, and yields interpretable modality-selection patterns that align with known radiological roles of T1, T1Gd, T2, and FLAIR [2], [8]. These findings support the effectiveness of task-aware multimodal fusion for robust adult glioma segmentation and are consistent with prior work on incomplete multimodal tumor segmentation [19], [24], [25].

V. CONCLUSION AND FUTURE WORK

In this study, we proposed RSMG, a region-specific modality gating framework for multimodal glioma segmentation. By learning separate modality policies for ET, TC, and WT within a hybrid CNN–Transformer backbone, RSMG reduces task interference and enables each sub-region to emphasize the most informative MRI sequences. Attention-guided and Grad-CAM-guided distillation with sparsity regularization further encourages stable and interpretable modality selection. On BraTS 2019, RSMG improves Dice from 0.901/0.916/0.909 to 0.914/0.928/0.921 and reduces HD95 from 5.41/7.26/5.88 mm to 4.98/6.72/5.43 mm, with consistent gains on BraTS 2021.

Future work will evaluate external benchmarks, stronger same-split baselines, computational cost, gate stability, and full missing-modality combinations. We will also explore how the model performs in real clinical settings.

REFERENCES

- [1] R. Onciul, F.-M. Brehar, C. Toader, et al., “Deciphering glioblastoma: Fundamental and novel insights into the biology and therapeutic strategies of gliomas,” *Current Issues in Molecular Biology*, vol. 46, pp. 2402–2443, 2024.
- [2] B. H. Menze, A. Jakab, S. Bauer, J. Kalpathy-Cramer, K. Farahani, J. Kirby, et al., “The multimodal brain tumor image segmentation benchmark (BRATS),” *IEEE Transactions on Medical Imaging*, vol. 34, no. 10, pp. 1993–2024, 2015.
- [3] Y. Yang, P. Wang, Z. Yang, et al., “Segmentation method of MRI brain tumor images based on improved U-Net network,” *Translational Cancer Research*, vol. 13, p. 1567, 2024.
- [4] O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional networks for biomedical image segmentation,” in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2015.
- [5] Ö. Çiçek, A. Abdulkadir, S. S. Lienkamp, T. Brox, and O. Ronneberger, “3D U-Net: Learning dense volumetric segmentation from sparse annotation,” in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2016.
- [6] J. Chen, Y. Lu, Q. Yu, X. Luo, E. Adeli, Y. Wang, L. Lu, A. L. Yuille, and Y. Zhou, “TransUNet: Transformers make strong encoders for medical image segmentation,” *arXiv preprint arXiv:2102.04306*, 2021.
- [7] Q. Pu, Z. Xi, S. Yin, Z. Zhao, and L. Zhao, “Advantages of Transformer and its application for medical image segmentation: A survey,” *BioMedical Engineering OnLine*, vol. 23, art. 14, 2024.
- [8] R. A. Zeineldin, M. E. Karar, Z. Elshaer, et al., “Explainable hybrid Vision Transformers and convolutional network for multimodal glioma segmentation in brain MRI,” *Scientific Reports*, vol. 14, art. 3713, 2024.
- [9] H. Ting and M. Liu, “Multimodal Transformer of Incomplete MRI Data for Brain Tumor Segmentation,” *IEEE Journal of Biomedical and Health Informatics*, vol. 28, no. 1, pp. 89–99, 2024.
- [10] Z. Zhang, G. Yang, Y. Zhang, H. Yue, A. Liu, Y. Ou, J. Gong, and X. Sun, “TMFormer: Token Merging Transformer for Brain Tumor Segmentation with Missing Modalities,” *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 7, pp. 7414–7422, 2024.
- [11] S. Karimijafarbigloo, R. Azad, A. Kazerouni, S. Ebadollahi, and D. Merhof, “MMCFFormer: Missing Modality Compensation Transformer for Brain Tumor Segmentation,” in *Medical Imaging with Deep Learning (MIDL), Proceedings of Machine Learning Research*, vol. 227, pp. 1144–1162, 2024.
- [12] Z. Sun, J. Li, Y. Wang, J. Cheng, Q. Zhou, and C. Li, “Unveiling Incomplete Modality Brain Tumor Segmentation: Leveraging Masked Predicted Auto-Encoder and Divergence Learning,” *arXiv preprint arXiv:2406.08634*, 2024.
- [13] Z. Zhang, Y. Lu, F. Ma, Y. Zhang, H. Yue, and X. Sun, “Incomplete Multi-modal Brain Tumor Segmentation via Learnable Sorting State Space Model,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 25982–25992, 2025.
- [14] J. Wang, L. Fan, W. Jing, D. Di, Y. Song, S. Liu, and C. Cong, “Hypergraph Tversky-Aware Domain Incremental Learning for Brain Tumor Segmentation with Missing Modalities,” in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pp. 283–293, 2025.
- [15] G. Liang, Q. Zhou, J. Chen, B. Huang, K. Chen, L. Gu, Z. Wang, S. Wu, and C. Yao, “Semantic-guided Masked Mutual Learning for Multi-modal Brain Tumor Segmentation with Arbitrary Missing Modalities,” *arXiv preprint arXiv:2507.07592*, 2025.
- [16] S. Zhu, Y. Chen, W. Chen, S. Jiang, G. Zhou, Y. Wang, F. Qin, C. Wang, and Q. Tian, “No Modality Left Behind: Adapting to Missing Modalities via Knowledge Distillation for Brain Tumor Segmentation,” *arXiv preprint arXiv:2509.15017*, 2025.
- [17] R. R. Selvaraju, M. Cogswell, A. Das, et al., “Grad-CAM: Visual explanations from deep networks via gradient-based localization,” in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pp. 618–626, 2017.
- [18] U. Baid, S. Ghodasara, M. Mohan, S. Bakas, et al., “The RSNA-ASNR-MICCAI BraTS 2021 benchmark on brain tumor segmentation and radiogenomic classification,” *arXiv preprint arXiv:2107.02314*, 2021.
- [19] Y. Ding, X. Yu, and Y. Yang, “RFNet: Region-Aware Fusion Network for Incomplete Multi-Modal Brain Tumor Segmentation,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 3955–3964, 2021.
- [20] Y. Zhang, N. He, J. Yang, Y. Li, D. Wei, Y. Huang, Y. Zhang, Z. He, and Y. Zheng, “mmFormer: Multimodal Medical Transformer for Incomplete Multimodal Learning of Brain Tumor Segmentation,” in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pp. 107–117, 2022.
- [21] E. Jang, S. Gu, and B. Poole, “Categorical Reparameterization with Gumbel-Softmax,” in *International Conference on Learning Representations (ICLR)*, 2017.
- [22] C. J. Maddison, A. Mnih, and Y. W. Teh, “The Concrete Distribution: A Continuous Relaxation of Discrete Random Variables,” in *International Conference on Learning Representations (ICLR)*, 2017.
- [23] G. Hinton, O. Vinyals, and J. Dean, “Distilling the Knowledge in a Neural Network,” *arXiv preprint arXiv:1503.02531*, 2015.
- [24] M. Havaei, N. Guizard, N. Chapados, and Y. Bengio, “HeMIS: Hetero-Modal Image Segmentation,” in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pp. 469–477, 2016.
- [25] C. Chen, Q. Dou, Y. Jin, H. Chen, J. Qin, and P.-A. Heng, “Robust Multimodal Brain Tumor Segmentation via Feature Disentanglement and Gated Fusion,” in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pp. 447–456, 2019.
- [26] G. Sun, Y. Pan, W. Kong, Z. Xu, J. Ma, T. Racharak, L.-M. Nguyen, and J. Xin, “DA-TransUNet: Integrating spatial and channel dual attention with transformer U-net for medical image segmentation,” *Frontiers in Bioengineering and Biotechnology*, vol. 12, art. 1398237, 2024.
- [27] L. Zhao, J. Ma, Y. Shao, C. Jia, J. Zhao, and H. Yuan, “MM-UNet: A multimodality brain tumor segmentation network in MRI images,” *Frontiers in Oncology*, vol. 12, art. 950706, 2022.
- [28] P. Ahmad, S. Qamar, L. Shen, S. Q. A. Rizvi, A. Ali, and G. Chetty, “MS UNet: Multi-scale 3D UNet for brain tumor segmentation,” in *Brainlesion: Glioma, Multiple Sclerosis, Stroke and Traumatic Brain Injuries*, Lecture Notes in Computer Science, vol. 12963, Springer, pp. 30–41, 2022.
- [29] D. Xiao, Q. Meng, S. Li, and X. Yuan, “Improving transformers with dynamically composable multi-head attention,” in *Proceedings of the 41st International Conference on Machine Learning*, Proceedings of Machine Learning Research, vol. 235, pp. 54300–54318, 2024.